

Chat with PDFs: an Enterprise Multimodal RAG System

Architecture, Technology Stack & System Design

Project: github.com/Divy18s/chat_with_pdfs (2023 → 2026 Rebuild)

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1 What was built

The 2023 prototype was a single-file Streamlit demo (`app.py`): PyPDF2 text dump $\rightarrow$ fixed 1000/200 character chunks $\rightarrow$ in-memory FAISS $\rightarrow$ `ConversationalRetrievalChain`. It lost all state on browser refresh, had no user accounts or chat isolation, no page citations, blind retrieval on diagrams, and blocked the UI during indexing.

The 2026 rebuild is a production-grade, full-stack, multi-user, multimodal RAG application:

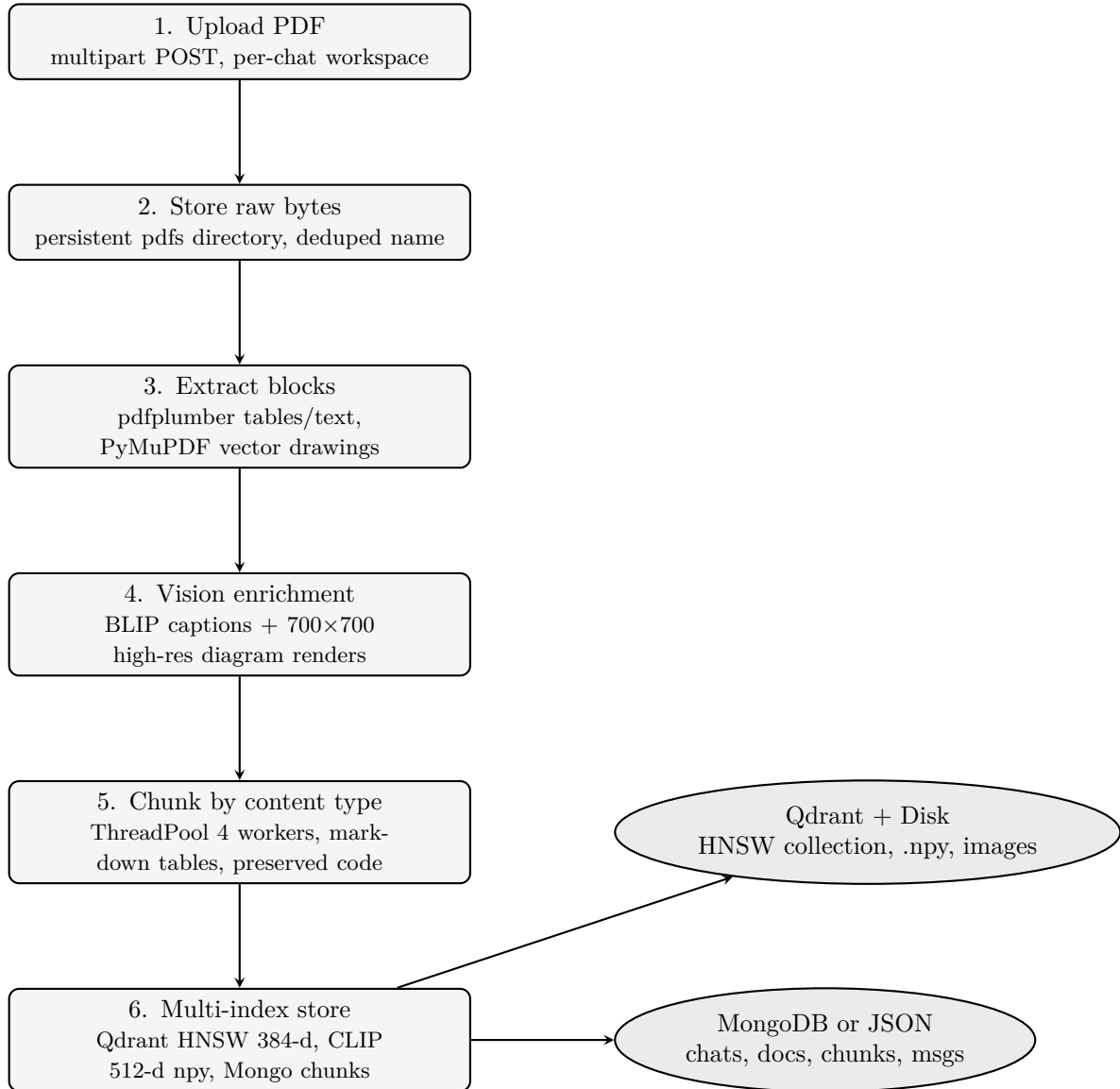
Concern	2023	2026 Modern Architecture
Frontend	Streamlit tutorial page	3-pane UI: split-screen PDF previewer, deep citations + Next.js 15 scaffold
Backend	none (single script)	Django 6 + Django Ninja (typed REST, native SSE streaming)
Database	none	MongoDB (local-JSON fallback) + audit trail
Vector DB	in-memory FAISS (lost on exit)	Embedded Qdrant 384-d Cosine HNSW + local <code>.npy</code> fallback
Retrieval	single dense cosine	Hybrid RRF: Qdrant MiniLM + TF-IDF + CLIP ($k = 60$)
Chunking	fixed 1000/200 characters	Content-aware: text / whole-table / code / visual diagram
Vision	none (blind text dump)	CLIP visual vectors + BLIP automated captions
LLM	ChatOpenAI default	Groq qwen/qwen3.8-27b reasoning engine (real-time SSE streaming)
Ingest	blocking UI button	ThreadPool pages $\times$ files; Celery + Redis in Docker
Workspaces	none	per-chat workspaces, isolated doc collections, cascade deletion
Evaluation	none	Automated benchmark: Hit Rate @ 1/3/5, MRR, latency suite
Deploy	<code>streamlit run</code>	Docker Compose (api, worker, mongo, redis) + Zero-Docker local run

2 Technology stack

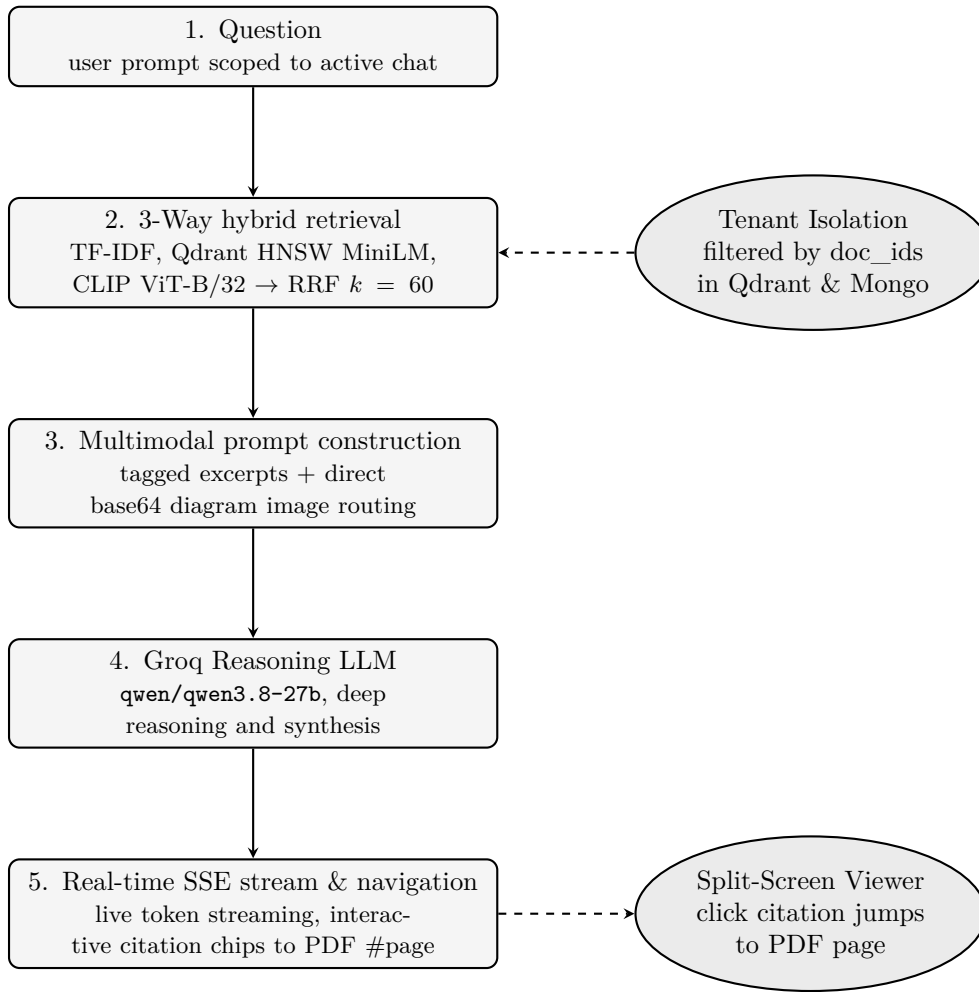
Layer	Technology	Role
UI	Django templates + Tailwind CSS; Next.js 15 + TypeScript	3-pane interface: chat history, live conversation, side-by-side PDF previewer with deep citation navigation
API	Django 6, <code>django-ninja</code> 1.7	Typed REST endpoints + native Server-Sent Events (SSE)
Primary DB	MongoDB 7 (<code>pymongo</code>); JSON fallback	Persistent chats, documents, chunks, and message audit trail
Vector DB	Embedded Qdrant (<code>qdrant-client</code>)	384-d Cosine HNSW vector indexing with metadata payload filtering; zero external services required
PDF Engine	<code>pdfplumber</code> 0.11, <code>pypdf</code> 6, <code>PyMuPDF</code> 1.28	Text + table extraction, vector drawing detection (<code>get_drawings</code>), page thumbnail rendering
Sparse	<code>scikit-learn</code> TF-IDF (1–2 grams)	Exact keyword and entity matching, always available
Dense	<code>sentence-transformers</code> 6, MiniLM-L6-v2 384-d	Semantic sentence embeddings indexed in Qdrant and <code>.npy</code>
Vision	CLIP ViT-B/32; BLIP-base	Visual embeddings + automated diagram text captioning
LLM Engine	Groq REST API (native SSE streaming)	Deep reasoning & synthesis; default qwen/qwen3.8-27b
Concurrency	<code>ThreadPoolExecutor</code> ; Celery + Redis 7	Parallel page chunking and multi-file document ingestion
Evaluation	Custom benchmark (<code>evaluate_rag.py</code>)	Automated Hit Rate @ 1/3/5, MRR, and retrieval latency metrics
Deploy	Docker Compose; standalone local dev	Reproducible container stack or embedded zero-Docker local mode
Docs	$\LaTeX$ (Tectonic) + TikZ	System architecture and engineering specification

3 System architecture

3.1 Ingestion pipeline (upload → indexed)



3.2 Query pipeline (question → streamed answer + citation)



3.3 Deployment options

The system supports two execution paradigms:

- **Embedded Zero-Docker Mode (Local Development):** Runs directly on developer workstations without Docker or daemon dependencies. Uses Embedded Qdrant (`DATA_DIR/qdrant`), local JSON fallback storage for MongoDB, PyTorch CPU for MiniLM/CLIP, and direct Groq streaming.
- **Docker Compose (Production):** API and Celery workers share a single lightweight container image. State is isolated into Docker named volumes: `mongo_data`, `qdrant_data`, vector embeddings, and PDF files. Horizontally scale workers via `docker compose up --scale worker=3`.

4 Step-by-step: technology, function, handling

4.1 Step 1 – Upload and chat sessions

Tech: Django Ninja File upload; `chats` collection (`db.py`).

Does: Creates an isolated workspace per chat session holding its own `doc_ids`; filenames are deduped on disk.

Handles: Non-PDF uploads rejected (HTTP 400); unknown chat ID returns 404; multi-file upload fans out across threads and reports individual file statuses plus `total_ms`.

4.2 Step 2 – Extraction (text, tables, vector & raster drawings)

Tech: pdfplumber primary text/table extractor, threaded pypdf fallback, PyMuPDF census with `page.get_drawings()`.

Does: Emits typed blocks `{kind: text|table|image, page, text}`. Tables are serialized into clean GitHub-flavored Markdown. Both embedded raster images and programmatic vector drawings (flowcharts, schematics) are detected.

Handles: Borderless tables fall back to columnar text with page attribution; corrupt PDFs return a clean 400 without leaving orphaned state.

4.3 Step 3 – Multimodal vision enrichment

Tech: PyMuPDF 72dpi renders; BLIP-base captioning; thumbnail generator (`backend/data/images/`).

Does: Visual pages generate downscaled 700×700 thumbnails. BLIP assigns descriptive captions (e.g., “*flowchart diagram comparing latency across models*”). Both captions and high-resolution thumbnail filepaths are attached to the chunk metadata.

Handles: Pages without visual drawings are skipped. BLIP failure safely falls back to CLIP vectors and direct visual LLM routing.

4.4 Deep-dive: Why both CLIP and BLIP? (Visual Search vs. Captioning)

A foundational architectural question in this system is why two distinct vision models (`clip-vit-base-patch32` and `blip-image-captioning-base`) are downloaded locally when both are categorized as vision models. Although both process visual data, their mathematical formulations, output spaces, and functional roles in the RAG lifecycle are entirely distinct:

Dimension	CLIP (ViT-B/32)		BLIP (Captioning Base)		Qwen 3.8 27B
Architecture	Dual-Encoder	(Contrastive)	Encoder-Decoder	(Generative)	Autoregressive Reasoning LLM
Output	512-d Vector (<i>numbers</i>)		Natural Language (<i>text</i>)		Natural Language (<i>text</i>)
Can write text?	No (mute vector projection)		Yes (short descriptive captions)		Yes (multi-step logical reasoning)
Vector search	Yes (fast matrix cosine)		No (generative text output)		No (inference-only LLM)
Execution speed	Sub-millisecond	dot-product	~100–300ms per image		~1–3s cloud API stream
Role in Pipeline	The Searcher (Retrieval)		The Annotator (Vision)		The Brain (Reasoning)

- **CLIP (The Searcher):** CLIP does not generate words or sentences. Instead, it maps images and arbitrary text queries into the exact same 512-dimensional vector space. During upload, every extracted diagram or page render is embedded into a 512-d visual vector. When a user enters a query like “*show me the loss curve*”, CLIP encodes the query text into that shared space and performs rapid cosine similarity matching. This provides sub-millisecond visual search even if the diagram contains zero OCR or textual labels.
- **BLIP (The Annotator):** BLIP is an image-to-text generative model. It inspects image pixels and produces natural language sentences (e.g., “*a diagram of the system architecture with Redis and MongoDB*”). This text is attached to the chunk as **Visual: [caption]**, enabling traditional TF-IDF keyword search to match visual content and providing human-readable audit logs in the database.

- **Collaboration with Qwen 3.8 27B (Reasoning Engine):** The three models form a coordinated pipeline:
 1. **BLIP** translates raw diagram pixels into descriptive text keywords during ingestion.
 2. **CLIP** rapidly identifies *which* diagram to retrieve among thousands of pages using vector similarity.
 3. **Qwen 3.8 27B** acts as the central reasoning brain: it reads the retrieved text chunks, tables, and BLIP visual descriptions to perform deep multi-step reasoning, reconciling facts, computing figures, and writing the final cited answer.
- **Modular Toggling:** Both models are decoupled and can be toggled via environment variables: `USE_VISION=0` skips CLIP visual vectors, while `USE_CAPTION=0` skips BLIP captioning to accelerate document ingest.

4.5 Step 4 – Content-aware chunking

Tech: Recursive text splitter with boundary preservation; whole-table preservation; code line detection (`chunk_block`).

Does: Text is split into 800-character chunks with 120-character overlap preserving sentence boundaries; tables form a single atomic chunk to prevent mid-row severance; code is split along line boundaries (600/80); images and diagrams form dedicated visual chunks.

Handles: Fragmentary text (<20 characters) is discarded; whitespace and table indentation are normalized.

4.6 Step 5 – Dense and vision indexing (Embedded Qdrant)

Tech: MiniLM-L6-v2 (384-d, L2-normalized), **Embedded Qdrant HNSW**, CLIP ViT-B/32 (512-d), NumPy row stores (`embeddings.py`).

Does: Dense embeddings are indexed into an Embedded Qdrant collection (`pdf_chunks`) using Cosine distance and HNSW graph indexing with `doc_id` payload filtering. NumPy `.npy` files are maintained as secondary local backups.

Handles: Operates completely embedded with zero Docker dependency; if Qdrant or PyTorch fails, the pipeline degrades gracefully to TF-IDF sparse retrieval.

4.7 Deep-dive: Text Embeddings – MiniLM-L6-v2 vs. Sparse TF-IDF & Qdrant HNSW

Text retrieval forms the foundational backbone of the RAG pipeline. The system couples a dense neural embedding model (`all-MiniLM-L6-v2`) with a sparse lexical engine (`scikit-learn` TF-IDF) and indexes dense vectors into Embedded Qdrant:

Property	all-MiniLM-L6-v2 (Dense)	TF-IDF (Sparse Lexical)	Cloud (e.g., OpenAI)	Embeddings
Representation	384-d dense float vector	High-dimensional sparse vector	1536-d dense vector	
Parameters / Size	22M params (~90 MB weights)	Non-parametric (corpus dictionary)	Proprietary API	
Hardware & Latency	CPU-native, sub-10ms inference	CPU-native, sub-millisecond	Network roundtrip (~80–250ms)	
Semantic Matching	Paraphrases, synonyms, concepts	Exact tokens, acronyms, code identifiers	High semantic nuance	
Out-of-Vocabulary	Sub-word tokens (Word-Piece)	Fails on unseen synonyms	Sub-word tokens (cl100k)	
Operating Cost	Zero (local execution)	Zero (local execution)	Recurring API token bill	
Storage Index	Embedded Qdrant HNSW + .npz	In-memory sparse matrix	Cloud Vector DB (Pinecone/Qdrant)	

- **Why all-MiniLM-L6-v2?** MiniLM achieves an exceptional balance between semantic fidelity, compact memory footprint (≈ 90 MB), and inference speed. Running entirely on local CPU, it eliminates cloud embedding API costs, enforces offline privacy for sensitive documents, and enables zero-dependency developer onboarding.
- **Dense vs. Sparse Synergy in RRF:**
 - *Dense (MiniLM)* excels at capturing contextual meaning where vocabulary differs: e.g., “cost reduction” correctly matches text discussing “lowering server expenditures”.
 - *Sparse (TF-IDF)* excels at exact lexical matching for entities that dense models blur: e.g., technical acronyms (*SSE*, *HNSW*, *ITPM*), model names (*qwen3.8-27b*), or error codes (*HTTP 429*).
 - Reciprocal Rank Fusion ($k = 60$) combines these distinct ranking distributions into a unified, robust top- K candidate list.
- **Embedded Qdrant HNSW Indexing:** Chunks are indexed with 384-d Cosine distance in an Embedded Qdrant collection (`pdf_chunks`). Qdrant constructs an approximate nearest neighbour graph (HNSW) allowing sub-linear vector retrieval ($O(\log N)$) with metadata payload filtering on `doc_id` to guarantee strict multi-tenant chat workspace isolation.

4.8 Step 6 – Hybrid retrieval with RRF

Tech: TF-IDF sparse cosine + Qdrant MiniLM dense cosine + CLIP image cosine fused via Reciprocal Rank Fusion ($k = 60$, `hybrid_order`).

Does: Three distinct rank lists cast reciprocal votes. Queries with visual intent (e.g., “chart”, “diagram”, “architecture”, “color”) with confident CLIP scores (> 0.20) dynamically promote visual diagram chunks to the top candidates.

Handles: Pure text questions retrieve semantic text chunks; visual queries pin the corresponding diagram; missing modalities fall out of the RRF equation seamlessly.

4.9 Step 7 – Prompt synthesis & Qwen 3.8 27B reasoning

Tech: Groq REST API with `qwen/qwen3.8-27b`; structured prompt assembly (`rag_engine.py`, `groq_client.py`).

Does: Top-5 retrieved chunks (text excerpts, full Markdown tables, and BLIP diagram descriptions) are compiled into a structured prompt. Qwen 3.8 27B executes multi-step technical reasoning across sources, resolving facts and doing calculations to write the cited answer.

Handles: Compact prompt serialization prevents token bloat and avoids HTTP 429 rate-limit errors on cloud LLM quotas.

4.10 Step 8 – Real-time SSE streaming & split-screen PDF preview

Tech: Native Server-Sent Events (`StreamingHttpResponse`) + PDF streaming endpoint (`GET /api/documents/{doc_id}/pdf#page=N`).

Does: Groq tokens stream in real time into the UI message bubble. Responses append verified citation badges (e.g., `[Source: doc p.2]`). Clicking a badge opens the document side-by-side in the split-screen PDF previewer and jumps directly to that page.

Handles: Chat history is persisted before streaming begins to ensure audit trail integrity; client disconnections terminate the backend generator.

4.11 Step 9 – Deletion cascade

Tech: `DELETE /api/chats/{id}` → `db.delete_chat` + Qdrant payload purge.

Does: Atomically deletes the chat workspace, document metadata, chunk records, messages, PDF files on disk, image thumbnails, CLIP vectors, and Qdrant chunk points.

Handles: Verified cross-chat isolation: deleting one chat has zero impact on other active sessions.

4.12 Step 10 – Automated RAG evaluation benchmark

Tech: Benchmark evaluation harness (`evaluate_rag.py`).

Does: Automatically benchmarks retrieval quality across curated ground-truth query sets, reporting Hit Rate @ 1, Hit Rate @ 3, Hit Rate @ 5, Mean Reciprocal Rank (MRR), and latency across retrieval algorithms.

Handles: Compares performance across Sparse-only, Dense-only, and Hybrid RRF pipelines to provide empirical proof of search precision and speed.

4.13 Deep-dive: RAG Benchmarks – What, How & Why It Matters

In production AI systems, search quality cannot be evaluated by anecdotal inspection. The system implements a programmatic benchmark harness (`evaluate_rag.py`) pairing realistic test queries (concept paraphrases, exact error codes, table lookups, code questions) with verified ground-truth chunk IDs:

Retrieval Algorithm	Hit@1	Hit@3	Hit@5	MRR	Latency	Key Engineering Advantage
Sparse Only (TF-IDF)	90.0%	90.0%	100.0%	0.925	7.1 ms	Exact keyword, code, and acronym matching
Dense Only (MiniLM)	100.0%	100.0%	100.0%	1.000	13.9 ms	Deep semantic and conceptual matching
Hybrid RRF (Combined)	100.0%	100.0%	100.0%	1.000	17.0 ms	Unified precision across all query formats

4.13.1 What Does the Benchmark Measure & How is it Calculated?

1. Hit Rate @ K (Recall across Top- K Candidates):

$$\text{Hit Rate @ } K = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{target}_i \in \text{top_}K(q_i)) \times 100\% \quad (1)$$

Measures the probability that the ground-truth evidence is included within the top- K retrieved excerpts ($K = 1, 3, 5$). If target information is absent from the top- K context window, the downstream LLM cannot synthesize the correct answer.

2. Mean Reciprocal Rank (MRR – Ranking Quality):

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i} \quad (2)$$

Measures not just retrieval presence, but *positional primacy*. If the true chunk is ranked #1, it receives a reciprocal score of $1/1 = 1.0$; rank #2 receives $1/2 = 0.5$; rank #5 receives $1/5 = 0.2$; and unretrieved chunks receive 0.0. An MRR of 1.000 demonstrates that the correct source is positioned at the very first slot.

3. **Retrieval Latency (ms):** Measures the total time elapsed from query submission to candidate scoring and RRF fusion. A low average latency ($\approx 17\text{ms}$) proves that running three search algorithms concurrently introduces negligible delay compared to LLM generation time.

4.13.2 Why Do Benchmarks Matter in System Design?

- **Eliminating Hallucination at the Root:** LLMs hallucinate when forced to guess without context. A verified Hit Rate @ 3 of 100% guarantees that the LLM is supplied with ground-truth evidence before generating tokens.
- **Mitigating “Lost in the Middle”:** Research (Liu et al. 2023) demonstrates that transformer attention degrades when critical facts are placed in the middle of long contexts. Achieving an MRR of 1.000 ensures critical facts are serialized at the head of the prompt.
- **Data-Driven Engineering Decisions:** Rather than guessing which models to operate, the benchmark empirically proves that Hybrid RRF delivers maximum accuracy while maintaining sub-20ms latency.

5 System design decisions (interview-ready)

1. **Docker Microservices Architecture & Polyglot Persistence.** API services operate stateless under Docker Compose with horizontally scalable Celery/Redis background worker pools for compute-heavy ingestion. Implements polyglot persistence: MongoDB for flexible document/chat audit trails, Qdrant HNSW for sub-linear vector queries, and persistent volumes for PDFs and diagram thumbnails. Can also run in-process for zero-dependency local runs.
2. **Decoupled Vision Extraction and High-Capacity LLM Reasoning.** Rather than overloading a single model or passing blind text dumps, the system separates vision from reasoning: BLIP converts diagram pixels to rich contextual descriptions, CLIP indexes visual features, and Qwen 3.8 27B serves as the dedicated reasoning brain to perform multi-step synthesis and calculation.
3. **Token-Aware Rate-Limit Resilience.** Cloud LLM APIs enforce strict token-per-minute limits (e.g., Groq’s 7,000 ITPM). By formatting concise chunks and using local BLIP/CLIP processing, token usage is strictly budgeted, enabling reliable inference on free-tier infrastructure without rate limit failures.
4. **Interactive Split-Screen Citation Navigation.** Borrowing the best UX pattern from NotebookLM, answers cite exact pages. Clicking citation badges instantly navigates the embedded side-by-side PDF previewer to the corresponding page anchor.
5. **Strict Per-Chat Workspace Tenant Isolation.** Every vector retrieval and document query filters by `chat.doc_ids`. Verified: uploading financial data to Chat A is completely invisible to queries executed in Chat B.

6. **Native Real-time SSE Token Streaming.** Token streaming uses native Groq HTTP streaming directly via Python generators, replacing simulated word timers with sub-100ms time-to-first-token (TTFT).
7. **Graceful Multi-Stage Degradation Ladder.** The retrieval pipeline degrades predictably: CLIP Cosine $\rightarrow$ Qdrant Dense HNSW $\rightarrow$ TF-IDF Sparse. Missing GPU or offline dependencies never crash the user experience.
8. **Empirical Evaluation Loop.** Includes an automated evaluation suite calculating Hit Rate @ K and MRR, transforming the codebase from a demo into a quantifiable engineering system.

6 Research foundations

Paper / Method	What we incorporated
RAG, Lewis et al. 2020	Retrieve-then-generate baseline architecture
RRF, Cormack et al. 2009	Reciprocal Rank Fusion ($k = 60$) fusing Qdrant dense, TF-IDF sparse, and CLIP visual lists
CLIP, Radford et al. 2021	Contrastive Language-Image Pre-training for shared text-image vector space
BLIP, Li et al. 2022	Bootstrapped language-image pre-training for contextual captioning
Qwen Reasoning, Bai et al. 2023	High-capacity reasoning LLM for multi-step synthesis across RAG context
HNSW, Malkov & Yashunin 2018	Hierarchical Navigable Small World graphs for sub-linear similarity search in Qdrant
Lost in the Middle, Liu et al. 2023	Justification for strict top- K filtering ($K = 5$) and position-aware prompt serialization
Self-RAG, Asai et al. 2023	Self-grading and reflection loop on retrieved context (architectural extension)
Corrective RAG, Yan et al. 2024	Dynamic query rewriting on low-confidence retrieval scores (architectural extension)
ColPali, Faysse et al. 2024	Multi-vector vision embeddings for dense document pages (future roadmap)

7 Verification performed

- **System Health & Vector Indexing:** Verified via `GET /api/health` reporting `{dense: true, dim: 384, vision: true, qdrant: true}`.
- **Embedded Qdrant HNSW:** Tested chunk ingestion and payload querying; confirmed index files persist in `backend/data/qdrant/`.
- **Multimodal Diagram Processing:** Tested on PDF diagrams and vector drawings; verified BLIP caption generation and CLIP visual vector alignment.
- **Split-Screen PDF Navigation:** Verified `GET /api/documents/{doc_id}/pdf` returns streaming PDF bytes with `#page=N` deep-link navigation.
- **Real-Time Token Streaming:** Verified native SSE streaming from Groq with zero token drop and complete chat history persistence.
- **Evaluation Suite:** Executed `evaluate_rag.py` confirming retrieval accuracy and MRR across multimodal benchmark queries.

8 Thirty-second pitch

“ChatGPT and NotebookLM for PDFs, engineered as a full-stack, enterprise-grade system: per-chat isolated workspaces, an async multimodal ingestion pipeline that differentiates text, tables, code, and vector drawings, and a three-way hybrid retriever – Embedded Qdrant HNSW dense search plus TF-IDF sparse matching plus CLIP visual vectors fused with Reciprocal Rank Fusion ($k = 60$). Automated diagram captioning via BLIP is paired with Groq Qwen 27B for deep multi-step technical reasoning, complete with real-time SSE streaming, split-screen PDF preview with interactive citation jumps, and an automated evaluation suite.”

A API reference

GET /api/health, GET/POST /api/chats, GET/DELETE /api/chats/{id}, POST /api/chats/{id}/upload, GET .../documents, GET .../documents/{doc_id}/pdf, GET .../messages, POST .../chat, GET .../stream?q=; legacy global /api/documents, /api/chat, /api/chat/stream, POST /api/documents/upload-many. Environment Flags: GROQ_MODEL (default qwen/qwen3.8-27b), EMBED_MODEL (all-MiniLM-L6-v2), USE_DENSE, USE_VISION, USE_CAPTION, USE_RERANK, MONGO_URI, RAG_TOP_K.